

SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation.

2 Mac Artifact Status

A deterministic synthetic S1 packet validates the readout format before real hybrid-model state dumps. It reports late/early max-abs ratio 8.461, std ratio 3.640, and kurtosis ratio 3.141, producing `SYNTHETIC_PASS_REAL_STATE_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets (granite-4.0-h-tiny, granite-4.0-h-small, granite-4.0-h-small-FP8, and Qwen3-Next-80B-A3B-Instruct) and ties them to shared architecture-map hashes, but all remain blocked for real S1 on this Mac because no corresponding model weights are cached repo-locally.

3 Next Evidence

The next admissible row is a real SSM-state dump from the smallest available hybrid reasoner. Promotion requires S1 heterogeneity on real states before any state-quantization recipe is tuned.

4 Admissible Real Packet

The real S1 packet must include model and tokenizer revisions, prompt manifest hash, seed list, context lengths, dtype, device, command, and the `architecture_map_hash` from the shared config-derived maps. Rows must report `model_id`, `model_revision`, `prompt_id`, `layer`, `position_bucket`, `state_shape`, `max_abs`, `rms`, `std`, `kurtosis`, `outlier_mass`, and a `bf16_no_quant` control. The packet is valid only if it passes `experimental.shared.check_gate_packet --mode real --project ssq_lr`. Saved tensor packets should be converted with `experimental.shared.hybrid_trace_packet_builder` before validation.

Required baseline	Purpose
BF16 no-op replay	Catch hook/serialization bugs.
INT8 state-only	Standard low-bit state control.
MXFP4 state-only	Candidate sub-FP16 recipe.
Random same-L2 noise	Separate structure from noise magnitude.
Scale shuffle	Test scale assignment sensitivity.
Byte accounting	Include scale/metadata overhead.

Table 1: Controls required before SSQ-LR can claim a state-quantization recipe.